6. Prove that

(i) $\frac{\cos 2A - \cos 3A}{\sin 2A + \sin 3A} = \tan\frac{A}{2}$

(ii) $\frac{\cos 7A + \cos 5A}{\sin 7A - \sin 5A} = \cot A$

7. Prove that

$\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

8. Evaluate

(i) $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$

(ii) $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ$

9. If $\cos A + \cos B = \frac{1}{2}$ and $\sin A + \sin B = \frac{1}{4}$, prove that $\tan\left(\frac{A+B}{2}\right) = \frac{1}{2}$

10. If $\sin(y+z-x)$, $\sin(z+x-y)$, $\sin(x+y-z)$, are in A.P, then prove that $\tan x$, $\tan y$ and $\tan z$ are in A.P

11. If $\operatorname{cosec} A + \sec A = \operatorname{cosec} B + \sec B$ prove that $\cot\left(\frac{A+B}{2}\right) = \tan A \tan B$

4.4 Inverse Trigonometric Functions

4.4.1 Inverse Trigonometric Functions

The quantities such as $\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$, etc., are known as inverse trigonometric functions.

i.e., if $\sin\theta = x$, then $\theta = \sin^{-1}x$. The domains and ranges (Principal value branches) of trigonometric functions are given below

Function	Domain x	Range y
$\sin^{-1}x$	$[-1, 1]$	$\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}x$	$R = (-\infty, \infty)$	$\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$
$\operatorname{cosec}^{-1}x$	$R-(-1, 1)$	$\left[\frac{-\pi}{2}, \frac{\pi}{2}\right], y \neq 0$
$\sec^{-1}x$	$R-(-1, 1)$	$[0, \pi], y \neq \frac{\pi}{2}$
$\cot^{-1}x$	R	$(0, \pi)$

4.4.2 Properties of Inverse Trigonometric Functions

Property (1)

(i) $\sin^{-1}(\sin x) = x$

(ii) $\cos^{-1}(\cos x) = x$

(iii) $\tan^{-1}(\tan x) = x$

(iv) $\cot^{-1}(\cot x) = x$

(v) $\sec^{-1}(\sec x) = x$

(vi) $\operatorname{cosec}^{-1}(\operatorname{cosec} x) = x$

Property (2)

(i) $\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1}(x)$

(ii) $\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1}(x)$

(iii) $\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1}(x)$

(iv) $\operatorname{cosec}^{-1}\left(\frac{1}{x}\right) = \sin^{-1}(x)$

(v) $\sec^{-1}\left(\frac{1}{x}\right) = \cos^{-1}(x)$

(vi) $\cot^{-1}\left(\frac{1}{x}\right) = \tan^{-1}(x)$

Property (3)

(i) $\sin^{-1}(-x) = -\sin^{-1}(x)$

(ii) $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$

(iii) $\tan^{-1}(-x) = -\tan^{-1}(x)$

(iv) $\cot^{-1}(-x) = -\cot^{-1}(x)$

(v) $\sec^{-1}(-x) = \pi - \sec^{-1}(x)$

(vi) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}(x)$

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